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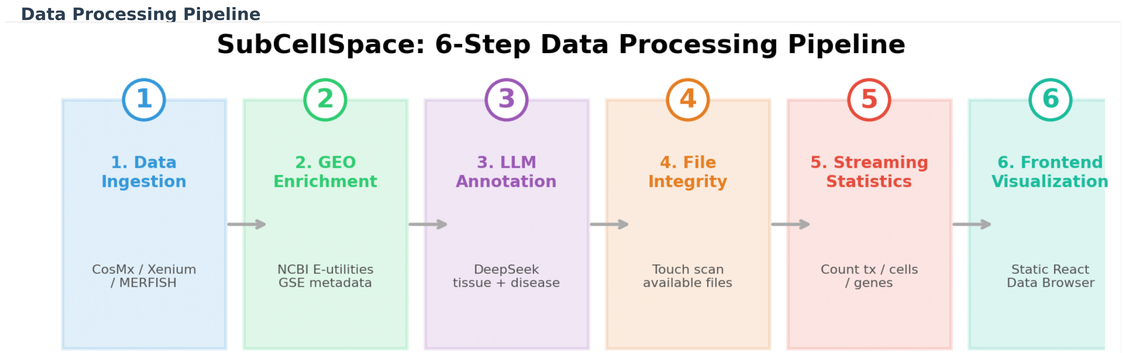
F. Li[1] · School of Life Sciences, Tsinghua University · 2026

Background

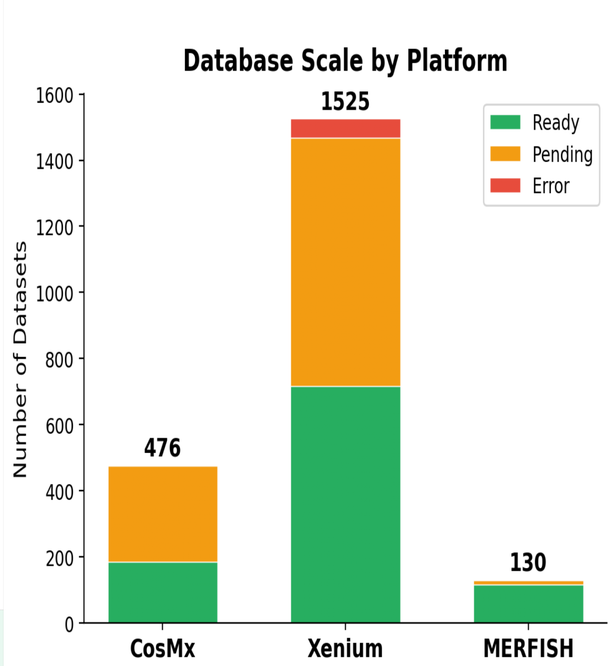
Spatial transcriptomics enables gene expression profiling within native tissue context. However, publicly available datasets are scattered across GEO, with inconsistent metadata and no unified platform for exploration. SubCellSpace addresses this by building a comprehensive, scverse-native database from three major platforms:

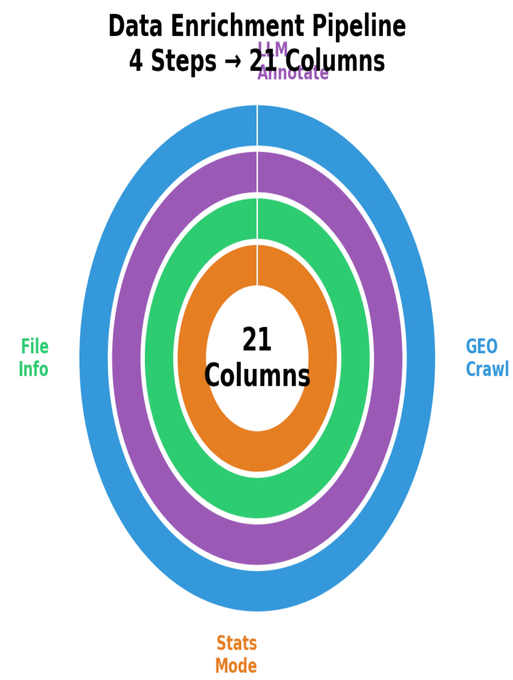
- NanoString CosMx — High-plex in situ RNA detection at single-cell resolution
- 10x Genomics Xenium — Subcellular transcriptomics with multi-modal imaging
- Vizgen MERFISH — Multiplexed FISH with ultra-high gene plexity

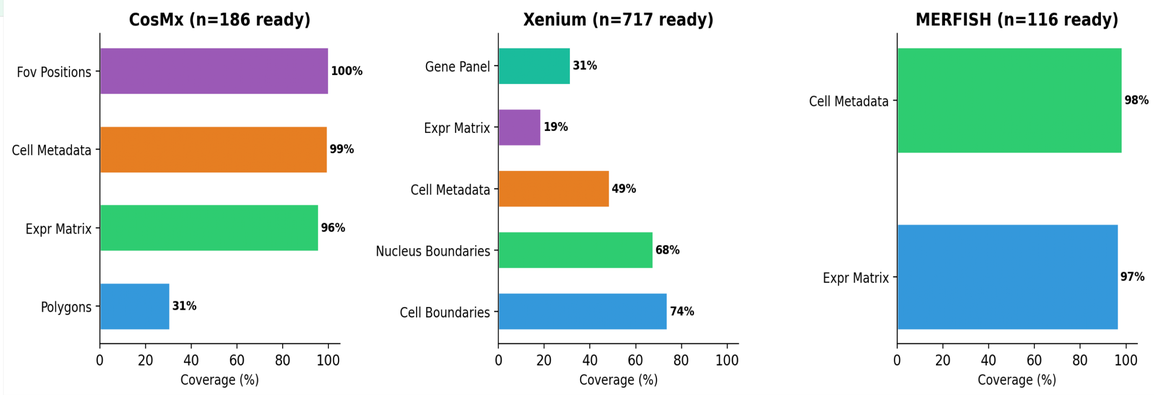
Data Processing Pipeline



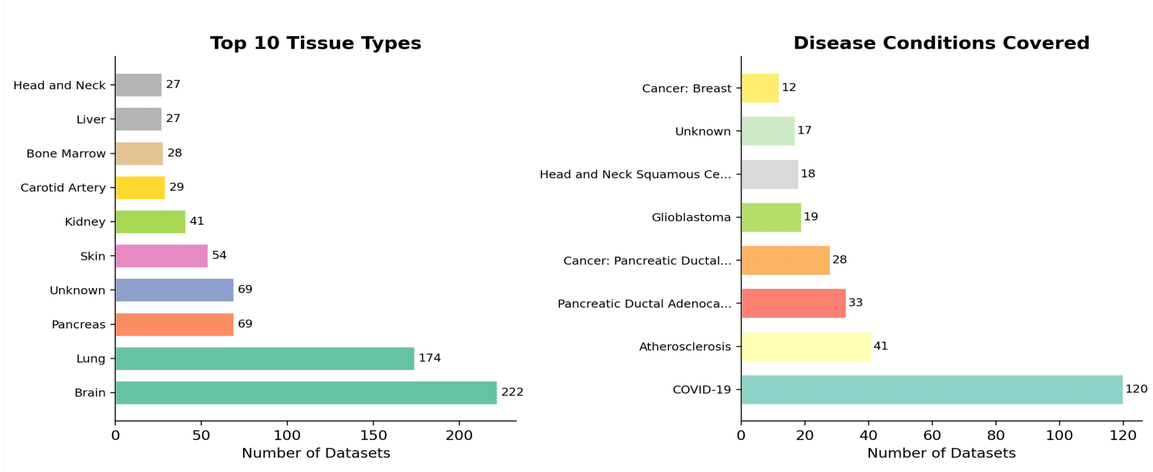
Database Scale







Biological Coverage



Tech Stack: Python · scanpy · squidpy · spatialdata · FastAPI · React · TypeScript · Slurm
Data Sources: NCBI GEO (E-utilities API) · DeepSeek LLM · 10x Genomics · NanoString/Bruker · Vizgen